

Problem Sheet 2 Solutions

1.

(a) If the supply is a voltage source V_s then the voltage at the load is found from the voltage divider rule

$$V_l = \frac{V_s R_l}{R_s + R_l}$$

Hence, the power in the load is

$$P_l = \frac{V_l^2}{R_l} = \frac{V_s^2 R_l}{(R_s + R_l)^2}$$

Simplifying then differentiating with respect to R_l gives

$$\frac{dP_l}{dR_l} = -\frac{V_s^2 \left(1 - \frac{R_s^2}{R_l^2}\right)}{\frac{R_s^2}{R_l} + 2R_s + R_l}$$

Which is zero for $R_s = R_l$.

(b) Total power is given by

$$P_t = \frac{V_s^2}{R_s + R_l}$$

Hence efficiency

$$\eta = \frac{P_l}{P_t} = \frac{R_l}{R_s + R_l}$$

2.

(a) Since the source is AC we need to use Root Mean-Square values to determine power

$$V_0 = 28 V_{\text{RMS}}$$

$$R_s = 500\Omega, R_L = 8\Omega \Rightarrow R_{\text{total}} = 508\Omega$$

$$P_{\text{RMS}} = V_{\text{RMS}}^2 R \Rightarrow P_s = 1.5\text{W}, P_l = 24 \text{ mW}$$

A clear case of a very poor impedance match: very little power is transferred from the source into the load. Efficiency is 1.6%.

(b) We'll add a signal transformer, which to a first approximation is a perfect transformer so the power in the primary winding is transformed into power out of the secondary winding with no loss. The Maximum Power Transfer Theorem tells us that in order to get the maximum power out of the source (amplifier) we need

$$Z_{\text{primary}} = R_s$$

Similarly, in order to get the maximum power out of the secondary winding into the load we need

$$Z_{\text{secondary}} = R_l$$

For a solenoid

$$Z = j\omega L$$

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$$L \propto n^2$$

The latter was given in lectures. The exact relation is

$$L = \frac{\mu_0 n^2 \pi r^2}{l}$$

where r is the solenoid radius and l is its length (the proof can be found in many E&M texts). This gives

$$Z_{\text{primary}} \propto n_1^2$$

$$Z_{\text{secondary}} \propto n_2^2$$

$$R_s \propto n_1^2, R_l \propto n_2^2$$

$$\frac{R_s}{R_l} = \frac{n_1^2}{n_2^2}$$

$$\frac{n_1}{n_2} = \sqrt{\frac{R_s}{R_l}}$$

When this condition is met, the combination impedance of transformer and the load R_l looks to the source like an impedance matched load $\equiv R_s$

(c) Hence now

$$P_s = \frac{V_0^2}{4R_s} = 392 \text{ mW}$$

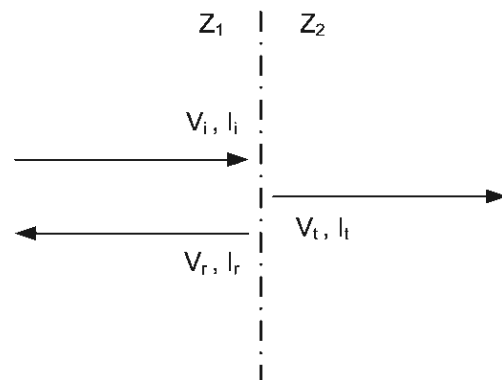
Which is now the same as the power driven into the load. (Note that the *total* power of the system has gone *down* due to the *decrease* in the overall impedance whilst the voltage remains constant). Efficiency is now 50%.

3.

(a) Conservation of charge tells us that either side of the interface there will be conservation of current (since current is charge per time). The charge going into the interface per time must equal the charge coming out of the interface in the same time. The interface is between two conductive media, so there can be no voltage discontinuity, hence the voltage at the interface is the same when viewed from either side. Denoting the instantaneous value, at the boundary, of the incident signal as V_i, I_i , the reflected signal as V_r, I_r and the transmitted as V_t, I_t we can write

$$V_i + V_r = V_t$$

$$I_i - I_r = I_t$$



The reflected current is negative since we interpret this as a signal travelling in the opposite direction to the incident signal.

(b) Since we know

$$V_i = I_i Z_1$$

$$V_r = I_r Z_1$$

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$$V_t = I_t Z_2$$

We can use the boundary conditions to get

$$\frac{V_i}{Z_1} - \frac{V_r}{Z_1} = \frac{V_t}{Z_2} = \frac{V_i + V_r}{Z_2}$$

Multiplying by $Z_1 Z_2$ and re-arranging

$$\frac{V_r}{V_i} = \frac{Z_2 - Z_1}{Z_2 + Z_1} = k_r$$

(c) Similarly, we can obtain

$$\frac{V_t}{V_i} = \frac{2Z_2}{Z_1 + Z_2} = k_t$$

4.

(a) The characteristic impedance is the same for all lengths of cable

$$Z_0 = \sqrt{\left(\frac{5 \times 10^{-7}}{100 \times 10^{-12}}\right)} = 70.7 \Omega$$

The propagation speed of the signal is

$$c_{coax} = \frac{1}{\sqrt{LC}} = 1.41 \times 10^8 \equiv 0.47 \times \text{light speed}$$

(b) If we set $C = 10$ pF/m then signal propagation speed would be 4.47×10^8 m/s which is greater than the speed of light and is hence un-physical. We can conclude that it is physically impossible to build a cable with such parameters.

(c) The amplitude reflection coefficient is given by

$$k_R = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

So

$$k_{out} = \frac{100 - 70.7}{100 + 70.7} = 0.172$$

Therefore $10 \times 0.172 = 1.72$ V is reflected. k_{out} is positive so the signal reflected from the output end of the cable is un-inverted.

(d) When this reflected signal reaches the input end it is reflected and inverted as k_{in} is negative.

$$k_{in} = \frac{10 - 70.7}{10 + 70.7} = -0.75$$

The pulse enters the cable at time $t = 0$. The transit time is

$$\frac{x_0}{\text{velocity}} = \frac{200}{1.41 \times 10^8} = 1.42 \times 10^{-6} \text{ s}$$

The first reflection at V_{out} has done 3 transits of the cable, the second has done 5.

Reflection 1

$$t = 4.26 \times 10^{-6} \text{ s}$$

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$$\text{Amplitude} = 10 \times 0.172 \times -0.75 = -1.29 \text{ V}$$

Reflection 2

$$t = 7.09 \times 10^{-6} \text{ s}$$

$$\text{Amplitude} = 10 \times 0.172^2 \times -0.75^2 = +0.17 \text{ V}$$

5.

(a) To achieve a minimum of reflection and maximum of transmission we should choose the intermediate section to have a characteristic impedance given by the geometric mean of the antenna and load impedances, i.e.

$$Z_0 = \sqrt{100 \times 50} = 70.7 \Omega$$

The section should be placed between the antenna and the 50Ω cable

(b) To perfectly match at 141MHz the cable needs to be $\lambda/4$ in length where λ is the wavelength of the signal in the matching section. From 4(a) we have the wave speed in 70.7Ω co-ax is $141 \times 10^6 \text{ ms}^{-1}$ hence $\lambda = 1\text{m}$ so the section should be 25cm in length

6.

(a) Assuming that the two Porro prisms are bonded together (not obvious from the diagram, but they usually are, as this avoids an unnecessary pair of air-glass boundaries), then we can count a total of 13 boundaries and classify them as follows:

5 air-to-convex

3 convex-to-concave

5 concave-to-air

In the lectures we saw that in a medium with relative permittivity ϵ and permeability μ

$$Z_{em} = \sqrt{\frac{\mu\mu_0}{\epsilon\epsilon_0}}$$

Also

$$c = \frac{1}{\sqrt{\epsilon\epsilon_0\mu\mu_0}}$$

With the refractive index n of the medium being the ratio of the speed of light in vacuum to the speed of light in the medium, hence

$$n = \sqrt{\epsilon\mu}$$

Since the relative permeability $\mu \cong 1$ for all but magnetic materials

$$n = \sqrt{\epsilon}$$

Therefore

$$Z \propto \frac{1}{n}$$

Also given in lectures was the reflection coefficient:

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$$k_R = \frac{Z_2 - Z_1}{Z_1 + Z_2}$$

Which we can rewrite as

$$k_R = \frac{1/n_2 - 1/n_1}{1/n_1 + 1/n_2}$$

Multiplying through by $n_1 n_2$

$$k_R = \frac{n_1 - n_2}{n_1 + n_2}$$

This is the amplitude reflection coefficient, which we need to square for light intensity which is proportional to power

$$I_R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

Power transmission coefficient is therefore

$$I_T = 1 - I_R = 1 - \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

Calculating for each of the three boundary types

Boundary	Type	n_1	n_2	I_T
A	Air/convex	1	1.75	0.9256
B	Convex/concave	1.75	1.52	0.9951
C	Concave/air	1.52	1	0.9574

Total transmission power is

$$I_T = I_{T_A}^5 \cdot I_{T_B}^3 \cdot I_{T_C}^5 \cong 54\%$$

(b)

Boundary	I_R	I_T
Air/convex	0.0744	0.9256
Air/MgF ₂	0.0255	0.9745
MgF ₂ /convex	0.0140	0.9860

For the combination Air/MgF₂/Convex we have

$$I_T = 0.9745 \times 0.9860$$

$$I_R = 1 - I_T = 0.0391$$

Hence reflection is reduced to

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$$\frac{0.0391}{0.0744} \cong 53\%$$

(c) Need to find $\frac{\lambda}{4}$ in the coating where the propagation speed is

$$c = \frac{c_0}{n_{\text{coating}}} = 2.17 \times 10^8 \text{ ms}^{-1}$$

$$\lambda/4 = 97 \text{ nm}$$

7.

(a) $Z_0 = \sqrt{\frac{L'}{C'}}$ where L' and C' are in henry and farad per metre respectively. Capacitors in parallel add together whereas two identical inductors in parallel have half the inductance (similar behaviour to two resistors in parallel). Hence characteristic impedance of the parallel combination

$$Z_{\parallel} = \sqrt{\frac{L/2}{2C}} = \frac{Z_0}{2}$$

In the case of a 50Ω cable, the parallel impedance is reduced to 25Ω .

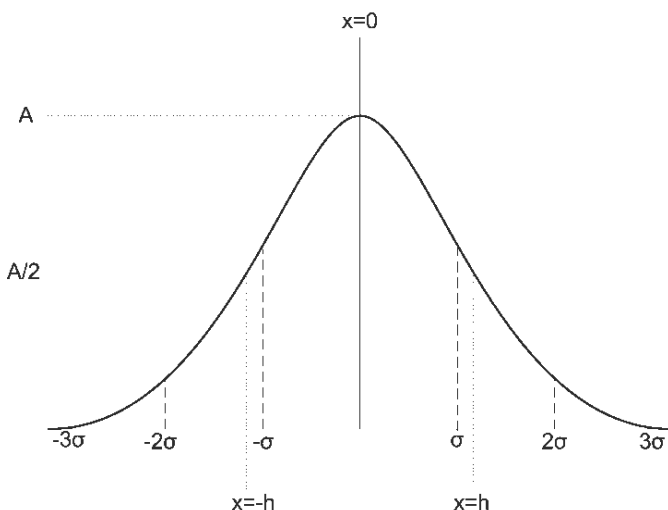
(b) Let $x = h$ at half the maximum height then

$$\frac{A}{2} = A e^{-\frac{x^2}{2\sigma^2}}$$

$$\ln \frac{1}{2} = -\frac{h^2}{2\sigma^2}$$

The full width at half maximum height is

$$2h = 2\sigma\sqrt{2 \ln 2} = 2.35\sigma$$



(c) The Gaussian is non-zero for all x . However, the cable is of finite length. So it is unphysical to say that the pulse is truly Gaussian. Furthermore, we could have grounded one end of the cable which will force the end to zero volts. Nevertheless, many short pulses generated by real instrumentation do have an *approximately* Gaussian shape.

8.

Instrumentation

(a) $Gain_{dB} = 20 \log_{10}(Gain)$, so at 100 Hz we have a gain of 10.

(b) The amplifier has an upper and lower -3 dB point. Reading off from the Bode plot we see that the two cut-off frequencies are at 10 and 1000 Hz. Hence the bandwidth is 990 Hz.

(c) At 1000 Hz we see that the gain is 17 dB. At 10000 Hz, 1 decade or multiple of 10 in frequency higher, the gain has fallen to 12 dB. Hence the roll-off is -5 dB/decade.

9.

(a) The 8-bit converter gives us $2^8 = 256$ digital numbers with which to measure the input range 0...1 volt. Hence the resolution of the measurement

$$R = \frac{1}{256} \text{ volt.}$$

If we assume that the digitizer rounds to the nearest digital value (in steps of R) then the voltages represented by 10 and 255 are $10R \pm (R/2)V$ and $255R \pm (R/2)V$, respectively. Other rounding schemes are possible, for example we could choose to round *down* to the nearest value of R in which case we would cover the full range of the input signal (see table right).

Voltage /V	Digital Value
0 to R	0
10R to 11R	10
255R to 1V	255

(b) $1/1000$ mV is then $0.26R$.

(c) The quantisation noise is the standard deviation of the uniform distribution between $-R/2$ and $+R/2$. This is $R/\sqrt{12}$ (given in lectures). Hence the quantisation noise is approximately $0.29R$. Since these are two independent random processes we can add their contribution in quadrature and the total noise is then

$$R\sqrt{(0.26^2 + 1/12)} = 0.39R$$

Quantisation has increased the noise by

$$\frac{0.39 - 0.26}{0.26} = 50\%.$$